

UNIT IV

STORAGE MANAGEMENT

Course Objectives: To classify and familiar with storage management and file systems

Course Outcomes: Analyze the File System Interface, Directory Structure.

Blooms Taxonomy Level: K1, K2, K3

Part-A

S.No.	Question	BTL *
1	State a file. A file is a named collection of related information that is recorded on secondary storage. A file contains either programs or data. A file has certain "structure" based on its type.	K1
2	List the various file attributes. A file has certain other attributes, which vary from one operating system to another, but typically consist of these: • Identifier, • Protection • Name, • Time • Type, • Date, • Location User identification and • Size	K1
3	Name the various file operations. The six basic file operations are: • Creating a file • Writing a file • Reading a file • Repositioning within a file • Deleting a file • Truncating a file	K1
4	List the different accessing methods of a file. Sequential Access: Data in the file is accessed in sequence, one record after another. Direct Access: Data can be accessed directly in any order without following a sequence. Indexed Sequential Access: An index is created for the file, which allows both sequential and direct access to records.	K1
5	Recall Directory. A directory is a structure in a file system that stores information about files, such as file name, location, size, and type. It acts like a symbol table that maps file names to their directory entries and helps in organizing and locating files on a disk.	K1
6	Recall the operations that can be performed on a directory. The operations that can be performed on a directory are • Search for a file • Create a file • Delete a file • Rename a file • List directory • Traverse the file system	K1

7	List the various layers of a file system. The file system is composed of many different levels. Each level in the design uses the feature of the lower levels to create new features for use by higher levels. - Application programs - Logical file system - File organization module - Basic file system - I/O control vi Devices	K1
8	Mention the structures used in file-system implementation. File system implementation uses two types of structures: 1. On-disk structures • Boot control block • Partition control block • Directory structure • File Control Block (FCB) 2. In-memory structures • In-memory partition table • In-memory directory structure • System-wide open file table • Per-process open file table	K1
9	Give the functions of virtual file system. • VFS separates file-system-independent operations from specific implementations by providing a common interface. • It allows transparent access to different types of file systems (local or network). • It uses a file representation called a vnode, and the kernel maintains a vnode for each active file or directory.	K1
10	Recall seek and latency time. The time taken by the head to move to the appropriate cylinder or track is called seek time. Once the head is at right track, it must wait until the desired block rotates under the read-write head. This delay is latency time.	K1
11	State the objectives of File Management System. • To organize and manage files efficiently in the system. • To store, retrieve, and update data in files easily. • To provide a hierarchical directory structure for organizing files. • To ensure protection and sharing of files among users.	K1
12	Name the content of a typical file control block. A File Control Block (FCB) is a file system data structure that stores information about a file and maintains the state of the file. The Contents of a typical File Control Block (FCB): • File permissions • File owner and group / ACL • File size • Pointers to file data blocks	K1
13	Mention the two types of system directories. Device directory, describing physical properties of files. File directory, giving logical properties of the files.	K1
14	Recall polling. Polling is the process where the computer waits for an external device to check for its readiness. The computer does not do anything else than checking the status of the device. Polling is often used with low-level hardware. Example: when a printer connected via a parallel port the computer waits until the next character has been received by the printer. These processes can be as minute as only reading 1 Byte. Polling is the continuous (or frequent) checking by a controlling device or process of other devices, processes, queues, etc.	K1
15	State buffering. A buffer is a memory area that stores data while they are transferred between two devices or between a device and an application. Buffering is done for three reasons:	K1

	a. To cope with a speed mismatch between the producer and consumer of a data stream b. To adapt between devices that have different data transfer sizes c. To support copy semantics for application I/O									
16	Distinguish between boot control block and partition control block. <table><tr><th>Boot Control Block</th><th>Partition Control Block</th></tr><tr><td>Contains information required to boot the operating system from the partition.</td><td>Contains details about the partition.</td></tr><tr><td>Usually the first block of the partition.</td><td>Stores information such as number of blocks, block size, free blocks, and FCB pointers.</td></tr><tr><td>If the disk has no operating system, this block may be empty.</td><td>Used by the file system to manage the partition.</td></tr></table>	Boot Control Block	Partition Control Block	Contains information required to boot the operating system from the partition.	Contains details about the partition.	Usually the first block of the partition.	Stores information such as number of blocks, block size, free blocks, and FCB pointers.	If the disk has no operating system, this block may be empty.	Used by the file system to manage the partition.	K2
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17	Recall spooling. A spool is a buffer that holds output for a device, such as printer, that cannot accept interleaved data streams. When an application finishes printing, the spooling system queues the corresponding spool file for output to the printer. The spooling system copies the queued spool files to the printer one at a time.	K1								
18	State rotational latency and disk bandwidth. Rotational latency is the additional time waiting for the disk to rotate the desired sector to the disk head. Disk bandwidth is the total number of bytes transferred, divided by the time between the first request for service and the completion of the last transfer.	K1								
19	Why disk scheduling is needed? Disk scheduling is needed to reduce the seek time of disk operations. Since many disk requests are placed in a queue, disk scheduling algorithms arrange the requests efficiently to minimize disk head movement, thereby improving system performance and reducing access time.	K2								
20	Name the characteristics that determines disk access speed. • Seek time • Rotational latency • Command processing time • Settle time	K1								

PART-B

S.No.	Question	BTL *
1	Illustrate FCFS, SSTF, C-SCAN and C-LOOK disk scheduling algorithms with examples. (16)	K2
2	Outline the various file allocation methods. (16)	K1
3	Write in detail about the Directory organization. (8)	K1
4	Elaborate two-level and acyclic graph schemes for defining the logical structure of a directory with an example. (16)	K2
5	On a disk with 200 cylinders, numbered 0 to 199, compute the number of tracks the disk arm must move to satisfy the entire request in the disk queue. Assume the last request received was at track 100. The queue in FIFO order contains requests for the following tracks. 55, 58, 39, 18, 90, 160, 150, 38, 184. Perform the computation to find the seek time for the all the disk scheduling algorithms. (16)	K3
6	Illustrate directory structure in detail. (8)	K2
7	Write a short note on free space management. (8)	K1
8	Case Study on college computer lab: (16) A college computer lab has a server with a large hard disk used to store files of students and teachers. Around 50 computers are connected to the server. Many users access, store, and share files at the same time. The operating system must manage the disk and organize files properly to ensure fast access, efficient storage, and data security. The system uses disk scheduling to reduce seek time when many disk requests occur. Files are stored using different access methods and organized using a hierarchical directory structure. External storage devices can be connected using file system mounting. Students working on group projects can share files, but file permissions are used for protection. The operating system also uses file allocation methods and free space management techniques to store files efficiently. Based on the above case study, illustrate the following: 1. Disk structure and disk scheduling 2. File concept and access methods 3. Directory structure and directory organization 4. File system mounting 5. File sharing and protection 6. File system implementation 7. Allocation methods and free space management.	K3